

Campus Resource Hub - Complete Workflow Guide

Project Overview

This project is a web application built using HTML, CSS, Python Flask, and SQLite3. It allows users to sign up, log in securely, upload PDF notes, search notes, and download notes. The system ensures secure password storage and prevents file conflicts using UUID-based filenames.

Technology Stack

Frontend: HTML, CSS
Backend: Python Flask
Database: SQLite3
File Storage: Local uploads folder
Security: Password hashing and strong password validation
File Safety: UUID-based unique filenames

Folder Structure

```
campus-resource-hub/  
  app.py  
  database.db  
  templates/  
    index.html  
    signup.html  
    login.html  
    upload.html  
    view.html  
  static/  
    style.css  
  uploads/
```

Database Design

Two main tables will be created:
Users table: stores username and hashed password.
Notes table: stores uploaded file details including UUID filename and uploader info.

User Authentication Workflow

User enters username and password.
Password strength is validated.
Password is hashed using secure algorithm.
Data is stored in SQLite3 database.
User logs in using credentials.

Session is created for user.

File Upload Workflow

User selects PDF file.
System generates UUID.
File is renamed using UUID.
File saved in uploads folder.
File info saved in database.
Conflict between same filenames is prevented.

View and Download Workflow

User opens view page.
Flask retrieves file list from database.
Files displayed in HTML page.
User clicks download.
Flask sends file from uploads folder.

Search Workflow

User enters search term.
Flask queries database using keyword.
Matching results displayed.

Security Measures

Passwords stored using hashing.
Strong password validation enforced.
Unique UUID filenames prevent overwrite.
Database prevents duplicate usernames.

Development Workflow

Step 1: Create folder structure
Step 2: Setup Flask app
Step 3: Setup database
Step 4: Implement signup and login
Step 5: Implement upload feature
Step 6: Implement view feature
Step 7: Implement download feature
Step 8: Implement search feature
Step 9: Test complete system
Step 10: Upload to GitHub

Final Result

A fully working Campus Resource Hub web application with secure login, file upload, unique file storage, search functionality, and download capability.